

Partial fractions integration

Basic skills

$$\text{Q1. a) } \frac{1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

$$\Rightarrow A(x+2) + B(x+1) = 1 \Rightarrow A+B=0, 2A+B=1$$

$$\text{so } A=1, B=-1$$

$$\text{so } \frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$$

$$\text{b) } \frac{4x^2}{(x+2)(x^2+16)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+16} \quad \leftarrow \text{All lower powers}$$

$$\Rightarrow A(x^2+16) + (Bx+C)(x+2) = 4x^2$$

$$\text{so } A+B=4, 2B+C=0, 16A+2C=0$$

$$\Rightarrow B=4-A \Rightarrow 8-2A+C=0 \Rightarrow 2A-C=8$$

$$\text{so } 20A=16 \Rightarrow A=\frac{4}{5} \Rightarrow B=\frac{16}{5}, C=-\frac{32}{5}$$

$$\text{so } \frac{4x^2}{(x+2)(x^2+16)} = \frac{4}{5(x+2)} + \frac{16x-32}{5(x^2+16)}$$

Q2. a) $\int \frac{1}{x^2+4} dx = \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right)$

$\underbrace{4}_{a^2} \rightarrow \int \frac{1}{x^2+a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$

b) $\int \frac{1}{\sqrt{81-x^2}} dx = \sin^{-1}\left(\frac{x}{9}\right)$

$\underbrace{81}_{a^2} \rightarrow \int \frac{1}{\sqrt{a^2-x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right)$

Q3. a) $x^3 - 5x^2 + 6x = x(x^2 - 5x + 6)$
 $= x(x-2)(x-3)$

b) $\frac{6}{x^3 - 5x^2 + 6x} = \frac{6}{x(x-2)(x-3)} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x-3}$

so $A(x-2)(x-3) + Bx(x-3) + Cx(x-2) = 6$

$\Rightarrow A+B+C=0, -5A-3B-2C=0, 6A=6$

$\Rightarrow A=1$ so $B+C=-1$ $3B+2C=-5$

$\Rightarrow C=-1-B \Rightarrow B-2=-5 \Rightarrow B=-7, C=6$

so $\frac{6}{x^3 - 5x^2 + 6x} = \frac{1}{x} - \frac{7}{x-2} + \frac{6}{x-3}$

$$c) \int \frac{6}{x^3 - 5x^2 + 6x} dx = \int \frac{1}{x} - \frac{7}{x-2} + \frac{6}{x-3} dx$$

$$= \ln x - 7 \ln(x-2) + 6 \ln(x-3)$$

$$Q4. \int \frac{1}{25 - x^2} dx = \frac{1}{2 \times 5} \ln \left| \frac{5+x}{5-x} \right|$$

$\underbrace{\quad}_{a^2} \rightarrow \int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{a+x}{a-x} \right|$

Competency skills

$$Q5. a) \int \frac{8}{(x^2+1)(x+1)} dx$$

$$\frac{8}{(x^2+1)(x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$

$$\Rightarrow A(x^2+1) + (Bx+C)(x+1) = 0 \Rightarrow \begin{matrix} A+B=0, \\ B+C=0 \\ A+C=8 \end{matrix}$$

$$\Rightarrow A = -B = C \text{ so } 2A = 8 \Rightarrow A = 4, B = -4, C = 4$$

$$\text{so } \int \frac{8}{(x^2+1)(x+1)} dx = \int \frac{4}{x+1} + \frac{4-4x}{x^2+1} dx$$

$$= \int \underbrace{\frac{4}{x+1}}_{\text{known}} + \underbrace{\frac{4}{x^2+1}}_{\text{formula}} - \underbrace{\frac{4x}{x^2+1}}_{\text{use}} dx$$

$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)|$

$$= \int \frac{4}{x+1} + \frac{4}{x^2+1} - 2 \frac{2x}{x^2+1} dx$$

$$= 4 \ln|x+1| + 4 \tan^{-1}(x) - 2 \ln|x^2+1| (+C)$$

$$b) \int \frac{x-6}{(x-1)(x^2+9)} dx, \quad \frac{x-6}{(x-1)(x^2+9)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+9}$$

$$\Rightarrow A(x^2+9) + (Bx+C)(x-1) = 0 \Rightarrow \begin{aligned} A+B &= 0, \\ C-B &= 1 \\ 9A-C &= -6 \end{aligned}$$

$$\text{So } A = -B \Rightarrow -9B - C = -6 \text{ so } 9B + C = 6$$

$$\text{So } C = 1+B \Rightarrow 10B + 1 = 6 \Rightarrow B = \frac{1}{2}$$

$$\text{So } A = -\frac{1}{2} \text{ and } C = \frac{3}{2}$$

$$\text{Therefore } \int \frac{x-6}{(x-1)(x^2+9)} dx = \frac{1}{2} \int \underbrace{\frac{-1}{x-1}}_{\substack{\text{natural} \\ \log}} + \underbrace{\frac{x}{x^2+9}}_{\substack{\int \frac{f'(x)}{f(x)}}} + \underbrace{\frac{3}{x^2+9}}_{\text{formula}} dx$$

$$= \frac{1}{2} \left(-\ln|x-1| + \frac{1}{2} \ln|x^2+9| + \frac{3}{3} \tan^{-1}\left(\frac{x}{3}\right) \right)$$

$$= \frac{1}{2} \left(\frac{1}{2} \ln|x^2+9| - \ln|x-1| + \tan^{-1}\left(\frac{x}{3}\right) \right)$$

Q6. $\int_3^4 \frac{8x+10}{(x^2+1)(x+4)} dx$

$$\frac{8x+10}{(x^2+1)(x+4)} = \frac{A}{x+4} + \frac{Bx+C}{x^2+1}$$

$$\Rightarrow A(x^2+1) + (Bx+C)(x+4) = 8x+10$$

$$\Rightarrow A+B=0, \quad 4B+C=8, \quad A+4C=10$$

$$\text{SO } B=-A \Rightarrow \begin{array}{r} -4A+C=8 \\ +4A+16C=40 \end{array} \quad \text{SO } 17C=48$$

$$\Rightarrow C = \frac{48}{17}, \quad A = 10 - 4 \times \frac{48}{17} = -\frac{22}{17}, \quad B = \frac{22}{17}$$

$$\text{SO } \int_3^4 \frac{8x+10}{(x^2+1)(x+4)} dx = \frac{1}{17} \int_3^4 \left(-\frac{22}{x+4} + \frac{22x}{x^2+1} + \frac{48}{x^2+1} \right) dx$$

$$= \frac{1}{17} \left[-22 \ln|x+4| + 11 \ln|x^2+1| + 48 \tan^{-1}(x) \right]_3^4$$

$$= \frac{1}{17} \left(-22 \ln 8 + 11 \ln 17 + 48 \tan^{-1} 4 + 22 \ln 7 - 11 \ln 10 - 48 \tan^{-1} 3 \right)$$

$$\approx 0.387$$

Q 7. $\int_2^3 \frac{1}{(x^2+5)(x-1)} dx$

$$\frac{1}{(x^2+5)(x-1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+5}$$

$$\Rightarrow A(x^2+5) + (Bx+C)(x-1) = 1 \Rightarrow \begin{matrix} A+B=0 \\ C-B=0 \\ 5A-C=1 \end{matrix}$$

$$C=B \Rightarrow \begin{matrix} +A+C=0 \\ 5A-C=1 \end{matrix} \Rightarrow 6A=1 \quad A=\frac{1}{6} \quad B=-\frac{1}{6} \quad C=-\frac{1}{6}$$

$$\text{So } \int_2^3 \frac{1}{(x^2+5)(x-1)} dx = \frac{1}{6} \int_2^3 \frac{1}{x-1} - \frac{x}{x^2+5} - \frac{1}{x^2+5} dx$$

$$= \frac{1}{6} \left[\ln|x-1| + \frac{1}{2} \ln|x^2+5| - \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{x}{\sqrt{5}}\right) \right]_2^3$$

$$= \frac{1}{6} \left(\ln 2 + \frac{1}{2} \ln 14 - \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{3}{\sqrt{5}}\right) - 0 - \frac{1}{2} \ln 9 + \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{2}{\sqrt{5}}\right) \right)$$

$$\approx 0.111$$

Q 8. a) $\int \frac{1}{x^2-4} dx = \frac{1}{2 \times 2} \ln \left| \frac{x-2}{x+2} \right| = \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right|$

or use

$\frac{1}{(x+2)(x-2)}$ to get result

$$\int \frac{1}{x^2-a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right|$$

b) Poly div :

$$\begin{array}{r} x^2 - 4 \\ x+1 \overline{) x^3 + x^2 - 4x - 4} \\ \underline{x^3 + x^2} \\ -4x - 4 \\ \underline{-4x - 4} \\ 0 \end{array}$$

so $x^3 + x^2 - 4x - 4 = (x+1)(x^2 - 4)$

c) $\int \frac{3}{x^3 + x^2 - 4x - 4} dx = \int \frac{3}{(x+1)(x^2 - 4)} dx$

$$\frac{3}{(x+1)(x^2 - 4)} = \frac{A}{x+1} + \frac{Bx+C}{x^2 - 4}$$

$$\Rightarrow A(x^2 - 4) + (Bx + C)(x+1) = 3 \Rightarrow \begin{array}{l} A+B=0, \\ B+C=0 \\ C-4A=3 \end{array}$$

so $B = -A = -C$ so $-3A = 3 \Rightarrow A = -1, B = 1, C = -1$

so $\int \frac{-1}{x+1} + \frac{x}{x^2 - 4} - \frac{1}{x^2 - 4} dx$

$$= -\ln|x+1| + \frac{1}{2} \ln|x^2 - 4| - \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right|$$

$$= \ln \frac{1}{|x+1|} + \ln(|x^2 - 4|^{\frac{1}{2}}) + \ln \left| \frac{x+2}{x-2} \right|^{\frac{1}{4}}$$

$$= \ln \left(\frac{|x^2-4|^{\frac{1}{2}} |x+2|^{\frac{1}{4}}}{|x+1| |x-2|^{\frac{1}{4}}} \right) + C$$

$$\text{For } 0 = \ln \left(\frac{|1-4|^{\frac{1}{2}} |2|^{\frac{1}{4}}}{|1| |1-2|^{\frac{1}{4}}} \right) + C$$

$$0 = \ln 2 + C \Rightarrow C = -\ln 2$$

$$\text{So } f(x) = \ln \left(\frac{|x^2-4|^{\frac{1}{2}} |x+2|^{\frac{1}{4}}}{|x+1| |x-2|^{\frac{1}{4}}} \right) - \ln 2$$

$$= \ln \left(\frac{|x^2-4|^{\frac{1}{2}} |x+2|^{\frac{1}{4}}}{2 |x+1| |x-2|^{\frac{1}{4}}} \right)$$

Advanced skills

$$\text{Q 9. a) } \frac{4kx}{(x-4k)(x^2+4k^2)} = \frac{A}{x-4k} + \frac{Bx+C}{x^2+4k^2}$$

$$\Rightarrow A(x^2+4k^2) + (Bx+C)(x-4k) = 4kx$$

$$\Rightarrow A+B=0, \quad C-4kB=4k, \quad 4k^2A-4kC=0$$

$$\text{So } B=-A \quad \text{So } C+4kA=4k, \quad 4k^2A=4k^2A$$

$$\Rightarrow A = \frac{C}{k} \quad \text{So } 5A=4k \quad A = \frac{4}{5}k, \quad B = -\frac{4}{5}k, \quad C = \frac{4}{5}k^2$$

$$\text{SO } \frac{4kx}{(x-4k)(x^2+4k^2)} = \frac{4}{5}k \left(\frac{1}{x-4k} - \frac{x}{x^2+4k^2} + \frac{k}{x^2+4k^2} \right)$$

$$\begin{aligned} \text{b) } & \int \frac{4}{5}k \left(\frac{1}{x-4k} - \frac{x}{x^2+4k^2} + \frac{k}{x^2+4k^2} \right) dx \\ &= \frac{4}{5}k \left(\ln |x-4k| - \frac{1}{2} \ln |x^2+4k^2| + k \times \frac{1}{2k} \tan^{-1} \left(\frac{x}{2k} \right) \right) \\ &= \frac{4}{5}k \left(\ln \left(\frac{|x-4k|}{\sqrt{x^2+4k^2}} \right) + \frac{1}{2} \tan^{-1} \left(\frac{x}{2k} \right) \right) \end{aligned}$$

$$\text{Q10. a) } \frac{2}{(1-x)(1+x)(1+x^2)} = \frac{A}{1-x} + \frac{B}{1+x} + \frac{Cx+D}{1+x^2}$$

$$\begin{aligned} \text{SO } & A(1+x)(1+x^2) + B(1-x)(1+x^2) + (Cx+D)(1-x)(1+x) = 2 \\ \Rightarrow & A(x^3+x^2+x+1) + B(-x^3+x^2-x+1) + (Cx+D)(-x^2+1) = 2 \\ \Rightarrow & A-B-C=0, \quad A+B-D=0, \quad A-B+C=0, \\ & A+B+D=2 \end{aligned}$$

$$\text{SO } C=0, \quad 2D=2 \text{ and } A=B \Rightarrow D=1$$

$$\Rightarrow A = \frac{1}{2}, \quad B = \frac{1}{2}, \quad C=0, \quad D=1$$

So $f(x) = \frac{1}{2(1-x)} + \frac{1}{2(1+x)} + \frac{1}{1+x^2}$

b) $\int_0^{\frac{1}{\sqrt{3}}} \frac{8}{1-x^4} dx$ as $f(x) = \frac{2}{1-x^4}$

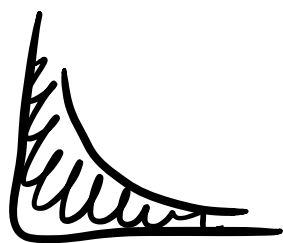
$\int_0^{\frac{1}{\sqrt{3}}} \frac{2}{1-x} + \frac{2}{1+x} + \frac{4}{1+x^2} dx$

$= \left[-2 \ln |1-x| + 2 \ln |1+x| + 4 \tan^{-1}(x) \right]_0^{\frac{1}{\sqrt{3}}}$

$= \left(-2 \ln \frac{2}{\sqrt{3}} + 2 \ln \frac{1+\sqrt{3}}{\sqrt{3}} + 4 \tan^{-1} \frac{1}{\sqrt{3}} - 0 \right)$

$= 2 \ln \left(\frac{\frac{1+\sqrt{3}}{\sqrt{3}}}{\frac{2}{\sqrt{3}}} \right) + 4 \times \frac{\pi}{6} = 2 \ln \left(\frac{1+\sqrt{3}}{2} \right) + \frac{2\pi}{3}$

Q11.



$\int_0^{\infty} \frac{1}{(x+1)(x^2+1)} dx$

need $\int_0^k \frac{1}{(x+1)(x^2+1)} dx$

$$\text{Now } \frac{1}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$

$$\Rightarrow A(x^2+1) + (Bx+C)(x+1) = 1 \Rightarrow \begin{matrix} A+B=0, \\ B+C=0, \\ A+C=1 \end{matrix}$$

$$\text{So } A = -B = C \quad \text{So } A = \frac{1}{2}, B = -\frac{1}{2}, C = \frac{1}{2}$$

$$\text{So } \frac{1}{2} \int_0^k \frac{1}{x+1} - \frac{x}{x^2+1} + \frac{1}{x^2+1} dx$$

$$= \frac{1}{2} \left[\ln|x+1| - \frac{1}{2} \ln|x^2+1| + \tan^{-1}(x) \right]_0^k$$

$$= \frac{1}{2} \left(\ln \frac{|k+1|}{\sqrt{k^2+1}} + \tan^{-1}(k) - 0 \right)$$

$$= \frac{1}{2} \left(\ln \left(\frac{1+\frac{1}{k}}{\sqrt{1+\frac{1}{k^2}}} \right) + \tan^{-1} k \right)$$

$$\text{As } k \rightarrow \infty \quad \frac{1}{2} \left(\ln \left(\frac{1+\frac{1}{k}}{\sqrt{1+\frac{1}{k^2}}} \right) + \tan^{-1}(k) \right)$$

$$\rightarrow \frac{1}{2} \left(\ln(1) + \frac{\pi}{2} \right) = \frac{\pi}{4}$$

$$\text{So Area is } \frac{\pi}{4}$$